HW7

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Q1: 3.8

Let X be a nonnegative random variable on some probability space.

We utilize the following results for (a) and (b):

- Theorem 3.1.1 (Markov's inequality): for nonnegative measurable f and $0 < t < \infty$, $\mu(\{f \geq t\}) \leq \frac{1}{t}\int f\,d\mu$.
- Proposition 3.1.8: if ϕ is twice differentiable on (a,b) and $\phi'' \geq 0$ on (a,b), then ϕ is convex on (a,b).
- Theorem 3.1.9 and Remark 3.1.2 (Jensen's inequality): if $P(X \in (a,b)) = 1$ and ϕ is convex on (a,b), then $\phi(EX) \leq E\phi(X)$, provided $E|X| < \infty$ and $E|\phi(X)| < \infty$.
- Remark 3.1.3: if ϕ is strictly convex (i.e., strict inequality holds in (1.2) for $0 < \lambda < 1$), then equality holds in Jensen's inequality iff $X = EX$ a.e. (P).
- Corollary 2.3.3: Proposition 2.3.1 holds for nonnegative measurable functions; in particular, $\int(\alpha f + \beta g)\,d\mu = \alpha\int f\,d\mu + \beta\int g\,d\mu$ for $\alpha,\beta \geq 0$, $f \geq g$ a.e. (μ) implies $\int f\,d\mu \geq \int g\,d\mu$, and $f = g$ a.e. (μ) implies $\int f\,d\mu = \int g\,d\mu$.
- Proposition 2.1.1(ii): if T_1 is $\langle \mathcal{F}_1, \mathcal{F}_2\rangle$-measurable and T_2 is $\langle \mathcal{F}_2, \mathcal{F}_3\rangle$-measurable, then $T_2 \circ T_1$ is $\langle \mathcal{F}_1, \mathcal{F}_3\rangle$-measurable.

(a)

Show that $(EX)\big(E\frac{1}{X}\big) \geq 1$. What does this say about the correlation between X and $\frac{1}{X}$?

Answer

$(EX)\big(E\frac{1}{X}\big) \geq 1$, so X and $\frac{1}{X}$ are negatively correlated whenever that correlation is defined.

Note: The inequality fails only when $X = 0$ a.e.: then $EX = 0$ and $E\frac{1}{X} = \infty$, and the product is 0 under the convention $0 \cdot \infty = 0$. If instead $P(X = 0) > 0$ but X is not 0 a.e., then $E\frac{1}{X} = \infty$ and $EX > 0$ by (1) below, so the product is $\infty \geq 1$. So we take $X > 0$ a.e. (P), so that $\frac{1}{X}$ is a.e. finite.

Proof.

Claim: For a random variable Y,

$$Y \geq 0 \text{ and } EY = 0 \implies Y = 0 \text{ a.e. } (P). \tag{1}$$

For each $k \geq 1$, Theorem 3.1.1 gives $P(Y \geq 1/k) \leq k\,EY = 0$. Since $\{Y > 0\} = \bigcup_{k\geq 1}\{Y \geq 1/k\}$, countable subadditivity gives $P(Y > 0) \leq \sum_{k\geq 1} P(Y \geq 1/k) = 0$.

Since $X > 0$ a.e., X is not 0 a.e., so $EX > 0$ by (1). Likewise $\frac{1}{X} > 0$ a.e., so $E\frac{1}{X} > 0$.

Case 1: $EX < \infty$ and $E\frac{1}{X} < \infty$. Define $\phi(x) \equiv \frac{1}{x}$ on $(0,\infty)$. Since $\phi''(x) = 2/x^3 > 0$, ϕ is convex (Proposition 3.1.8). Since $P(X \in (0,\infty)) = 1$, $E|X| = EX < \infty$, and $E|\phi(X)| = E\frac{1}{X} < \infty$, Jensen's inequality gives

$$\frac{1}{EX} = \phi(EX) \leq E\phi(X) = E\frac{1}{X}.$$

Multiplying by $EX \in (0,\infty)$, $(EX)\big(E\frac{1}{X}\big) \geq 1$.

Case 2: $EX = \infty$ or $E\frac{1}{X} = \infty$. Since both are positive, $(EX)\big(E\frac{1}{X}\big) = \infty \geq 1$.

In either case, $(EX)\big(E\frac{1}{X}\big) \geq 1$.

Correlation: For the correlation to be defined, we need $EX^2 < \infty$, $E\frac{1}{X^2} < \infty$, and $\mathrm{Var}(X), \mathrm{Var}\big(\frac{1}{X}\big) > 0$. Since $x \leq 1 + x^2$, this puts us in Case 1. Since $X \cdot \frac{1}{X} = 1$,

$$\mathrm{Corr}\Big(X, \frac{1}{X}\Big) = \frac{E\big(X \cdot \frac{1}{X}\big) - (EX)\big(E\frac{1}{X}\big)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}\big(\frac{1}{X}\big)}} = \frac{1 - (EX)\big(E\frac{1}{X}\big)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}\big(\frac{1}{X}\big)}} \leq 0.$$

The inequality is in fact strict. Since $\phi'' > 0$, the mean value theorem argument behind Proposition 3.1.8 gives strict inequality in (1.2), so ϕ is strictly convex. Equality in Jensen's inequality would then force $X = EX$ a.e. (P) (Remark 3.1.3), and then $\mathrm{Var}(X) = 0$. //

Said differently, X and $\frac{1}{X}$ are negatively correlated unless X is a constant, in which case the correlation is undefined.

(b)

Let $f, g : \mathbb{R}_+ \to \mathbb{R}_+$ be Borel measurable and such that $f(x)g(x) \geq 1$ for all x in $\mathbb{R}_+$. Show that $Ef(X)Eg(X) \geq 1$.

Answer

Proof. Since $f, g \geq 0$ and $f(x)g(x) \geq 1$, neither f nor g vanishes on $\mathbb{R}_+$. So

$$f(x) \geq \frac{1}{g(x)} > 0 \quad \text{for all } x \in \mathbb{R}_+.$$

Define $Y \equiv g(X)$. Since g is Borel measurable and X is a random variable, Y is a random variable by Proposition 2.1.1(ii), and $Y > 0$ everywhere. Note that $f(X) \geq \frac{1}{Y}$ pointwise, so $Ef(X) \geq E\frac{1}{Y}$ by Corollary 2.3.3. Also $EY > 0$ by (1).

Putting this all together, and applying (a) to Y,

$$Ef(X)\,Eg(X) \geq \Big(E\frac{1}{Y}\Big)(EY) \geq 1. \quad //$$

Unlike (a), this needs no assumption that $X > 0$, since $g > 0$ on all of $\mathbb{R}_+$.

Q2: 3.9

Prove Corollary 3.1.13 using Hölder's inequality applied to an appropriate measure space.

Answer

Recall Corollary 3.1.13: Let $k \in \mathbb{N}$. Let $a_1, \ldots, a_k, b_1, \ldots, b_k$ be real numbers and $c_1, \ldots, c_k$ be positive real numbers.

(i) Then, for any $1 < p < \infty$,

$$\sum_{i=1}^{k} |a_i b_i| c_i \leq \left(\sum_{i=1}^{k} |a_i|^p c_i\right)^{\frac{1}{p}} \left(\sum_{i=1}^{k} |b_i|^q c_i\right)^{\frac{1}{q}}, \tag{1.16}$$

where $q = \frac{p}{p-1}$.

(ii)

$$\sum_{i=1}^{k} |a_i b_i| c_i \le \left(\sum_{i=1}^{k} |a_i|^2 c_i \right)^{\frac{1}{2}} \left(\sum_{i=1}^{k} |b_i|^2 c_i \right)^{\frac{1}{2}}. \tag{1.17}$$

Note: The book uses $q = \frac{p-1}{p}$ in (i); but here we (I) use the conjugate exponent $q = \frac{p}{p-1}$ of Theorem 3.1.11, such that $\frac{1}{p} + \frac{1}{q} = 1$.

We utilize the following results:

- Theorem 3.1.11 (Hölder's inequality): for $1 < p < \infty$, $f \in L^p(\Omega, \mathcal{F}, \mu)$ and $g \in L^q(\Omega, \mathcal{F}, \mu)$ with $q = \frac{p}{p-1}$, $\int |fg| \, d\mu \le \left(\int |f|^p \, d\mu \right)^{\frac{1}{p}} \left(\int |g|^q \, d\mu \right)^{\frac{1}{q}}$.
- $L^p(\mu)$ (Section 2.5): the measurable f with $\int |f|^p \, d\mu < \infty$.
- Remark 2.3.2: on a countable $\Omega = \{\omega_i : i \ge 1\}$ with $\mathcal{F} = \mathcal{P}(\Omega)$, every $h : \Omega \to [0, \infty)$ is measurable, and $\int h \, d\mu = \sum_i h(\omega_i) \, \mu(\{\omega_i\})$.

Proof. Define the measure space $(\Omega, \mathcal{F}, \mu)$ by

$$\Omega \equiv \{1, 2, \ldots, k\}, \qquad \mathcal{F} \equiv \mathcal{P}(\Omega), \qquad \mu(A) \equiv \sum_{i \in A} c_i, \quad A \subset \Omega.$$

Note that μ is a measure. $\mu(\emptyset) = 0$ and $\mu \ge 0$, since each $c_i > 0$. For disjoint $A_1, A_2, \ldots \in \mathcal{F}$, each $i \in \bigcup_{n \ge 1} A_n$ lies in exactly one A_n, so

$$\mu\Big(\bigcup_{n \ge 1} A_n \Big) = \sum_{n \ge 1} \sum_{i \in A_n} c_i = \sum_{n \ge 1} \mu(A_n).$$

That is, μ is the counting measure on Ω with weight $\mu(\{i\}) = c_i$ on each point.

Define $f, g : \Omega \to \mathbb{R}$ by $f(i) \equiv a_i$ and $g(i) \equiv b_i$. Since $\mathcal{F}$ contains every subset of Ω, every function on Ω is $\mathcal{F}$-measurable. By Remark 2.3.2, for any $1 < p < \infty$ and $q \equiv \frac{p}{p-1}$,

$$\int |f|^p \, d\mu = \sum_{i=1}^{k} |a_i|^p c_i, \qquad \int |g|^q \, d\mu = \sum_{i=1}^{k} |b_i|^q c_i, \qquad \int |fg| \, d\mu = \sum_{i=1}^{k} |a_i b_i| c_i. \tag{2}$$

Each is a finite sum of finite terms, so $f \in L^p(\mu)$ and $g \in L^q(\mu)$.

(i): Let $1 < p < \infty$ be given, and let $q \equiv \frac{p}{p-1}$. By Hölder's inequality and (2),

$$\sum_{i=1}^{k} |a_i b_i| c_i = \int |fg| \, d\mu \le \Big(\int |f|^p \, d\mu \Big)^{\frac{1}{p}} \Big(\int |g|^q \, d\mu \Big)^{\frac{1}{q}} = \left(\sum_{i=1}^{k} |a_i|^p c_i \right)^{\frac{1}{p}} \left(\sum_{i=1}^{k} |b_i|^q c_i \right)^{\frac{1}{q}},$$

which is (1.16).

(ii): Taking $p = 2$ in (i) gives $q = \frac{2}{2-1} = 2$, and (1.16) becomes (1.17). //

Q3: 3.11

Verify Minkowski's inequality for $p = 1$ and $p = \infty$.

Answer

We utilize the following results:

- Section 3.2.1 (L^p norms): for $0 < p < \infty$, $\|f\|_p = \left(\int |f|^p \, d\mu \right)^{\min\{1/p, 1\}}$, so $\|f\|_1 = \int |f| \, d\mu$; and $\|f\|_\infty \equiv \sup\{k : \mu(\{|f| > k\}) > 0\}$ (the essential supremum).

- Theorem 3.1.14 (Minkowski's inequality): for $1 < p < \infty$ and $f, g \in L^p(\mu)$, $\|f+g\|_p \le \|f\|_p + \|g\|_p$.
- Proposition 3.2.1: for $0 < p \le \infty$, $f, g \in L^p(\mu)$ and $a, b \in \mathbb{R}$ imply $af + bg \in L^p(\mu)$.
- Corollary 2.3.3 (see Q1).
- Monotonicity and countable subadditivity of measures (Chapter 1): $A \subset B$ implies $\mu(A) \le \mu(B)$, and $\mu\big(\bigcup_n A_n\big) \le \sum_n \mu(A_n)$.

Proof. Let $p \in \{1, \infty\}$ and $f, g \in L^p(\mu)$ be given. By Proposition 3.2.1 (with $a = b = 1$), $f + g \in L^p(\mu)$, so $\|f+g\|_p$ is defined. For every $\omega \in \Omega$, the triangle inequality on $\mathbb{R}$ gives

$$|f(\omega) + g(\omega)| \le |f(\omega)| + |g(\omega)|.$$

$p = 1$: Since $|f+g|$, $|f|$, and $|g|$ are nonnegative and measurable, Corollary 2.3.3 gives

$$\|f+g\|_1 = \int |f+g|\, d\mu \le \int \big(|f| + |g|\big)\, d\mu = \int |f|\, d\mu + \int |g|\, d\mu = \|f\|_1 + \|g\|_1.$$

$p = \infty$: We first show that, off a null set, $|h|$ is bounded by its essential supremum.

Claim: For $h \in L^\infty(\mu)$, $\mu(\{|h| > \|h\|_\infty\}) = 0$.

Write $M \equiv \|h\|_\infty$, which is finite since $h \in L^\infty(\mu)$ (with $\sup \emptyset \equiv 0$, which arises only when $\mu(\Omega) = 0$). Let $k > M$ be given. If $\mu(\{|h| > k\}) > 0$, then k lies in the set defining M, so $k \le M$, a contradiction. Hence $\mu(\{|h| > k\}) = 0$ for every $k > M$. Since

$$\{|h| > M\} = \bigcup_{n \ge 1} \{|h| > M + n^{-1}\},$$

countable subadditivity gives $\mu(\{|h| > M\}) \le \sum_{n \ge 1} \mu(\{|h| > M + n^{-1}\}) = 0$, which proves the Claim.

Now let $N \equiv \{|f| > \|f\|_\infty\} \cup \{|g| > \|g\|_\infty\}$. By the Claim and subadditivity, $\mu(N) = 0$. For $\omega \in N^c$,

$$|f(\omega) + g(\omega)| \le |f(\omega)| + |g(\omega)| \le \|f\|_\infty + \|g\|_\infty,$$

so $\{|f+g| > \|f\|_\infty + \|g\|_\infty\} \subset N$. Let $k \ge \|f\|_\infty + \|g\|_\infty$ be given. Then $\{|f+g| > k\} \subset N$, so by monotonicity $\mu(\{|f+g| > k\}) = 0$. Hence no such k lies in $\{k : \mu(\{|f+g| > k\}) > 0\}$, and every element of that set is less than $\|f\|_\infty + \|g\|_\infty$. Taking the supremum,

$$\|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty.$$

Combining both cases, Minkowski's inequality holds for $p = 1$ and $p = \infty$, as asserted in Remark 3.1.5. //

Q4: 3.17 (a) and (b)

Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mu)$. Recall that $Eh(X) = \int h(X)\, d\mu$ if $h(X) \in L^1(\Omega, \mathcal{F}, \mu)$.

We utilize the following results for (a) and (b):

- Theorem 3.1.11 (Hölder's inequality; see Q2), together with its equality case: if $\|fg\|_1 \ne 0$, then equality holds in (1.13) iff $|f|^p = c|g|^q$ a.e. (μ) for some constant $c \in (0, \infty)$.
- Corollary 2.3.3 (see Q1).
- The Claim (1) in Q1, with μ in place of P.

(a)

Show that $(E|X|^{p_1}) \le (E|X|^{p_2})^{\frac{p_1}{p_2}}$ for any $0 < p_1 < p_2 < \infty$.

Answer

Proof. Let $0 < p_1 < p_2 < \infty$ be given. If $E|X|^{p_2} = \infty$, the right side is ∞ and there is nothing to show, so suppose $E|X|^{p_2} < \infty$.

Define $Y \equiv |X|^{p_1}$, $r \equiv \frac{p_2}{p_1} > 1$, and $s \equiv \frac{r}{r-1}$, so that $\frac{1}{r} + \frac{1}{s} = 1$. Note that $Y^r = |X|^{p_2}$, so $\int Y^r \, d\mu = E|X|^{p_2} < \infty$ and $Y \in L^r(\mu)$. Also $\int 1^s \, d\mu = \mu(\Omega) = 1$, so $1 \in L^s(\mu)$. By Hölder's inequality with $f = Y$ and $g = 1$,

$$E|X|^{p_1} = \int |Y \cdot 1| \, d\mu \leq \Big(\int Y^r \, d\mu \Big)^{\frac{1}{r}} \Big(\int 1^s \, d\mu \Big)^{\frac{1}{s}} = (E|X|^{p_2})^{\frac{p_1}{p_2}}. \quad // \tag{3}$$

(b)

Show that '=' holds in (a) iff $|X|$ is a constant a.e. (μ).

Answer

Note: This requires $E|X|^{p_2} < \infty$. Otherwise, any X with $E|X|^{p_1} = \infty$ gives $\infty = \infty$ in (a), constant or not.

Proof. Let Y, r, and s be as in (a), with $E|X|^{p_2} < \infty$.

($\Leftarrow$) Suppose $|X| = c$ a.e. (μ) for some constant $c \geq 0$. Then by Corollary 2.3.3, $E|X|^{p_1} = c^{p_1}$ and

$$(E|X|^{p_2})^{\frac{p_1}{p_2}} = (c^{p_2})^{\frac{p_1}{p_2}} = c^{p_1},$$

so equality holds.

($\Rightarrow$) Suppose equality holds in (3).

Case 1: $E|X|^{p_1} = 0$. Then $|X|^{p_1} = 0$ a.e. (μ) by (1), i.e., $|X| = 0$ a.e. (μ), a constant.

Case 2: $E|X|^{p_1} > 0$. Then $\|Y \cdot 1\|_1 = E|X|^{p_1} \neq 0$, so the equality case of Hölder's inequality gives a constant $c' \in (0, \infty)$ with $|Y|^r = c' \cdot 1^s$ a.e. (μ), i.e., $|X|^{p_2} = c'$ a.e. (μ). Thus $|X| = (c')^{1/p_2}$ a.e. (μ), a constant.

In either case, $|X|$ is a constant a.e. (μ). //

Q5: 3.18

Let X be a nonnegative random variable.

We utilize the following results for (a) and (b):

- Proposition 3.1.8 (see Q1).
- Theorem 3.1.9 and Remark 3.1.2 (Jensen's inequality; see Q1).
- Corollary 2.3.3 (see Q1).
- The Claim (1) in Q1.

(a)

Show that $EX \log X \geq (EX)(E \log X)$.

Answer

Note: We take both sides to be well defined and finite: $X > 0$ a.e. (P), so that $\log X$ is a.e. finite, and $E|X \log X|$, $E|\log X| < \infty$. (Then $EX < \infty$ as well, since $X \leq e + |X \log X|$.)

Proof. By (1), $EX > 0$, so $EX \in (0, \infty)$ and $\log(EX)$ is finite. Both steps below apply Jensen's inequality on $(0, \infty)$, since $P(X \in (0, \infty)) = 1$.

Convexity of $x \log x$: Define $\phi(x) \equiv x \log x$ on $(0, \infty)$. Since $\phi''(x) = 1/x > 0$, ϕ is convex (Proposition 3.1.8). Since $E|X| < \infty$ and $E|\phi(X)| = E|X \log X| < \infty$, Jensen's inequality gives

$$(EX) \log(EX) = \phi(EX) \leq E\phi(X) = E(X \log X).$$

Concavity of $\log x$: Define $\psi(x) \equiv -\log x$ on $(0, \infty)$. Since $\psi''(x) = 1/x^2 > 0$, ψ is convex (Proposition 3.1.8). Since $E|\psi(X)| = E|\log X| < \infty$, Jensen's inequality gives $-\log(EX) \leq -E \log X$, i.e., $E \log X \leq \log(EX)$. Multiplying by $EX > 0$,

$$(EX)(E \log X) \leq (EX) \log(EX).$$

Putting this all together,

$$E(X \log X) \geq (EX) \log(EX) \geq (EX)(E \log X). \quad //$$

(b)

Show that $\sqrt{1 + (EX)^2} \leq E(\sqrt{1 + X^2}) \leq 1 + EX$.

Answer

Proof. Define $\phi(x) \equiv \sqrt{1 + x^2}$, $x \in \mathbb{R}$.

Upper bound: For $x \geq 0$, $1 + x^2 \leq 1 + 2x + x^2 = (1 + x)^2$, so

$$\phi(x) \leq 1 + x \quad \text{for all } x \geq 0. \tag{4}$$

Since $X \geq 0$, (4) gives $\phi(X) \leq 1 + X$ pointwise, and so by Corollary 2.3.3,

$$E\sqrt{1 + X^2} \leq E(1 + X) = 1 + EX.$$

Lower bound:

Case 1: $EX = \infty$. Then $\sqrt{1 + (EX)^2} = \infty$. Since $\phi(x) \geq x$, Corollary 2.3.3 gives $E\phi(X) \geq EX = \infty$ as well, so the lower bound reads $\infty \leq \infty$.

Case 2: $EX < \infty$. Since $\phi'(x) = x(1 + x^2)^{-1/2}$ and $\phi''(x) = (1 + x^2)^{-3/2} > 0$, ϕ is convex on $\mathbb{R}$ (Proposition 3.1.8). Also $E|X| = EX < \infty$, and $E|\phi(X)| \leq 1 + EX < \infty$ by the upper bound. Jensen's inequality on $(-\infty, \infty)$ then gives

$$\sqrt{1 + (EX)^2} = \phi(EX) \leq E\phi(X) = E\sqrt{1 + X^2}.$$

Putting everything together, $\sqrt{1 + (EX)^2} \leq E(\sqrt{1 + X^2}) \leq 1 + EX$. //